Category	Sub-category	Code Name	Definition
Feature Use	Left Panel	Event View	Code only when user clicks the button to enter the events view. Do not code subsequent rows while the user remains here.
		Devices View	Code only when user clicks the button to enter the devices view. Do not code subsequent rows while the user remains here.
		IP Address Filter	Adds a filter from the left panel IP Address filter bar
		Event Type Filter	Adds a filter from the left panel Event Type filter bar
		Tags & Star Filter	Adds a filter from the left panel Tags/Stars filter bar
		Add Data Export (Left Panel)	
		Compare Events	Uses "pin for comparison"
		Delete Data Export	
	Events View	Edit Columns	
		Chronological Sort	
		Time Filter	
		Search Bar	
		Add/Remove Tag	
		Add/Remove Star	
		Click Event Row	Clicks an expandable event row within the table
		Add Data Export (Events View)	
		Enable/Disable/Delete Filter Chip	Toggles the visibility of a filter chip or removes it
		Temp Enable/Disable Data Export	
		Event Row Filter	Adds a filter using the Filter icon on each attribute row within the expanded event row
	Devices View	Rename/Annotate Device	Adds a nickname or tag to a device row
		Click Device Details	Expanding an accordion with device info (instance or profile)
		See Events (Devices View)	Uses "See events" button attached to a given events group
		IP Query	Uses the IP address hyperlink in the Devices dropdown to see events with a given IP address
		Session ID Query	Uses the session cookie hyperlink in the Devices dropdown to see events with that cookie
	Home & Misc	Explore my data	Button on the home screen
		Home page button	
		Data ingestion flow	Code only when user clicks the button to start the data ingestion flow and select a platform. Do not code subsequent rows while the user remains here.
		Interactive Demo	Code only when user clicks the button to enter the demo. Do not code subsequent rows while the user remains here.
		Skip Demo	
		How to Request	Code only when user clicks the button to enter the page. Do not code subsequent rows while the user remains here.
		Github/Research Button	
		Safe Exit	
		Help Modal	Generic tag for any inline help icon or popup window, includes masked UA.
Initiation of Feature Use *(Apply this code to the same line as feature use, not the interviewer's prompt.)*		Prompted - Unsolicited	Interviewer prompted participant to use a feature (the participant was not necessarily looking for it)
		During demo	Feature was used in the demo (considered unprompted if used later since this is an affordance)
		Prompted to aid task	The participant was looking for a feature and the interviewer pointed them to it or suggested it might solve their task
		Unprompted	Participant used feature naturally without being prompted or told
Participant Experience/Reaction with Feature	General Feedback	Positive Feedback	Expresses approval or satisfaction.
		Negative Feedback	Expresses general dislike or frustration with existing functionality.
		Mixed Feedback	Expresses partial approval of a specific feature, or is not sure if they like it. Do not confuse this with people being polite about negative feedback.
	Gradations of "confusion"	Confusion -- UI	Expresses confusion about what a UI feature does or clearly hesitates.
		Confusion -- Data	Expresses confusion about the underlying data.
		Mistake	User makes a (non-bug) mistake while navigating or uses a button accidentally. "Oops"
		Bug	Reports broken functionality or errors.
		Expectation mismatch	What the user expects from the feature does not align with what it does
		Clarifying Question	Asks about how a feature works or what a term means.
	Suggestions	Text Suggestion	Proposes rewording an existing label or term.
		UI Suggestion	Proposes changing an existing visual element or layout.
		New Feature	Proposes adding a non-existent tool, view, or capability.
STRUCTURAL CODES	***Structural subcodes***	***Inductive subcodes***	***Definition/Notes***
Investigation Practices	Task start (only for bookkeeping purposes)	Task 1	
		Task 2	
		Task 3	
		Task 4	
		Task 5	
	Hypothesis/Questions	Dates device was logged in	Speculating about the dates a specific device was or was not logged-in
		Method of authentication	Speculating about how the adversary obtained access in the ground truth scenario (recovery flow, login, shared device)
		Activity attribution	Speculating about who or which device was the source of some event or set of events (i.e., a session, an IP address, etc)
		Identity of unknown device	A hypothesis about whether "iphone" is alice's or bob's device, or a third altogether.
	Actions	Looking at temporally proximate events	Reasons about hypothesis by looking at the events within a certain time window (i.e., rapid succession of logins = compromise)
		Correlating IP addresses to activity	Reasons about hypothesis by which events or devices share an IP address.
		Comparing/looking at device or session details	Reasons about hypothesis using session cookie, device IDs (model, OS, user agent).

Category	Sub-category	Code Name	Definition
		Correlating activity across 2+ platforms	Reasons about hypothesis about platform A by using an existing hypotheses or fact about platform B (often co-occurring with IP addr).
		Searches Internet	Searches internet to aid hypothesis
		Takes notes	Takes notes on an external
		Reference ASI	References the ASI either as physical phone OR the screenshots in the reference materials
	Conclusion	Task answer	Give full or partial answer to task or hypothesis. Can dually-coded with hypothesis (a hypothesis may be a a sufficient task answer)
		Self-corrects hypothesis	Walks back initial hypothesis or adds a significant caveat ("actually, wait, no...,")
		Interviewer corrects/confirms hypothesis	Interviewer jumps in with a full or partial task explanation, usually to move the study along
Uncertainty about Data	**Data limitations** uncertainty about how the platform has captured and recorded user actions	Missing/incorrect/unstable device identifiers	The underlying data export is missing or has incorrect device IDs
		Recorded events	Uncertainty or confusion about how the platform records different authentication events (recovery flows, logins).
		Incorrect measurement (time, location)	Uncertainty about whether the platform's recorded timestamps and locations (and other metrics) are correct
		Length or time-limit of log	Uncertainty about how far back in time different files in the data export go.
	Client's context uncertainty about how the adversary might have accessed network, devices, accounts, etc. (that may not necessarily be represented in the data itself)	Shared home/devices	Caveats or questions about whether the hypothetical client and adversary shared a home or whether they physically shared devices
		Spoofing or VPN use	Caveats about technologically spoofing device IDs, like user agents; or use of VPNs that modify origin IP address
		Recovery Methods	Caveats or uncertainty about what recovery methods (email, pwd code, etc) the adversary could have accessed in the hypothetical scenario
	LEStrADE's interpretation uncertainty about how the the tool has represented or transformed data	Event taxonomy	User is uncertain about how tool is translating or representing types of events ("logins vs authentication challenges")
		Event-device relationships	User is uncertain about how different devices relate to different events *specifically because of the interface or how querying/hierarchy works in the tool*
		Device attributes	User is uncertain about how lestrade is parsing and joining device IDs (user agents, IPs, session cookies, etc)
		Timestamps	User is uncertain about how lestrade is communicating the significance of different time stamps
Friction & Suggestions	Common points of friction	Filters	Annoyances or UI friction related to addition/deletion/disablement of filter chips in the events view
		Event noise	High volume of events or irrelevant events obscuring the events the user wants to look at
		Mentally organizing strings/dates	*User has trouble remembering or mentally organizing strings (IPs, user agent strings) or dates*
		Difficulty in mapping investigation task to specific feature	User struggles to figure out which UI features would help them complete the task
		Too much guidance	User notes that tool is providing too much nudging/handholding/guidance or automated interpretation
		Difficulties comparing events/devices	Difficulty with comparing 2+ devices or events. This might be use of several tabs, noting how annoying it is to scroll, etc [1]
	Suggestions/New Features	Comparison view / multi-tab	User wants to be able to diff the values in the table containing event or device details
		Activity timeline by device	User wants to be able to see an event timeline where multiple devices are represented in different rows/columns --> "interleaving"
		Glossary	User suggests annotating/defining jargon (i.e., user agent)
		Add manual/external events	User suggests the ability to add manual events in the timeline (e.g., when Bob moved out)
		Most recent events	(Keeping this for now, might subsume into another code eventually)
		Progressive disclosure in event/device details	User suggests using hierarchy or progressive disclosure to make device or event details (expanded) easier to read
		Graph view	A view (graphical, tabular, etc) that relates different devices or sessions to the origin IP address
		IP augmentation	Geolocation a certain IP address, determining if IP addresses are from VPNs, letting users annotate IP addresses,
		Nudges/help/tour	User wants more tutorials/nudges/help modals. Includes wanting infra to build hypotheses and have them be corrected if obviously inconsistent with data
		Interviewer suggests new feature	Interviewer brings up an idea in development or was suggested previously to get feedback (separate natural suggestions from promed
		Additional Annotation	User wants more expansive or robust annotation features in the UI
		Additional Filtering	User wants more options for filters or ways to stack filters
		Reset/Export	User wants an option to reset or export all annotations.
		Re-order visual elements	User wants to re-order visual elements on a page, generally.
Deployment Considerations	In Tech Clinic	Limited time during consultations	Points out logistical difficulty of using LEStrADE during the short time window of a typical consultation
		Requesting data exports	Considerations about requesting data exports in the clinic setting; which categories; how to request; time between request and receipt
		Overwhelm / high cognitive load	User mentions that the amount of information might overwhelm the client or consultant
		Client communication	Difficulties or strategies they would use when communicating info to client, including **disclaimers about certainty, corroboration client context, asking before taking a decision, Distracts from attending to client, teaching the client about the data exports**
		Re-traumatizing to client	Points out that data contents or volume could be stressful or overwhelming to client
		Benefits	Remarks about how or why the tool might be beneficial specifically in the clinic setting, how the tool might address a difficulty users frequency encounter
		Real user data	Considerations related to the size/scale of real user data; privacy considerations about collaboratively working with real user data
	Learnability	Jargon & Terminology	Anticipated issues about how non-techncial users might struggle with jargon or technical concepts
		Wants/can't find existing feature	Participant expresses a desire for or suggests a feature that already exists (since they have not yet found it)
		Easier with future use	Participant mentions that the tool may become easier to use in the future after they use it more
		(Non) Interest in Tour	Remarks about learning the tool through the tour OR stated noninterest in the tour feature
	Trust & Reliability	Testing, bugs, and compatibility	How bugs impact trust, need for testing, browser compatibility, etc
		Developer reputability	Remarks about trustworthiness and privacy motivations of the developer.
		Communicating tool's purpose/scope	Remarks about how to communicate what the tool does (parse) and does not/cannot do (classify). What limitations arise from the tool vs the data at a high level (i.e., in tutorials, landing page)
		Distinguishing between parsed data & augmentation/inference	Somewhat similar to the above; remarks about how the tool communicates the *difference* between individual data records. Which fields were augmented or normalized vs which come from the original dataset? How can the analyst verify?